

SEAT NO. SE-24009

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
FIRST YEAR (Software)
FALL SEMESTER EXAMINATIONS 2024

Time : 3 Hours

Batch 2024

Dated : 09-DEC-24
 Max Marks : 60

Calculus - MT-173

NOTE: Attempt all questions. All questions carry equal marks.

Q. No. 1

Evaluate the following integrals:

$$\int_0^1 dz \int_0^2 dx \int_0^x (x + y + z) dy$$

[CLO-01] (06)

Q. No. 2

If $z^3 - 3yz - 3x = 0$, then prove that:

$$z \frac{\partial z}{\partial x} = \frac{\partial z}{\partial y}$$

[CLO-01] (06)

Q. No. 3

State and prove the De Moivre's theorem for all integers 'n'.

[CLO-01] (06)

Q. No. 4

Draw the graph of the function:

$$f(x) = \begin{cases} -x & x < 0 \\ x^2 & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

[CLO-01] (06)

Q. No. 5

Using Gamma functions, evaluate:

$$\int_0^{\infty} e^{-x^2} x^4 dx$$

[CLO-01] (06)

P.T.O

$$\sin x = \cos x$$

Q. No. 6

Find the locus of a point in the plane satisfying the following conditions:

(i) $|z - 5| = 6$

(ii) $\operatorname{Re}(iz) = 3$

[CLO-02] (06)

Q. No. 7

Show that at any point on the equiangular spiral, $r = a e^{\theta \cot \alpha}$; the radius of curvature is $\rho = r \operatorname{cosec} \alpha$, where α is a constant.

[CLO-02] (06)

Q. No. 8

Examine the absolute maximum and minimum values of $f(x, y) = x^2 + y^2 + 6x + 12$ on a triangular plate in the first quadrant bounded by the lines $x = 0$, $y = 0$, and $y = 9 - x$.

[CLO-02] (06)

Q. No. 9

Find a root of $\cos x + e^x - 4 = 0$ correct up to four decimal places by Newton-Raphson method.

[CLO-02] (06)

Q. No. 10

Find the percentage error in the volume $V = xyz$ of a parallelepiped if z decreases by 2 % (-0.02) and x, y increase by 0.75 % (0.0075).

$\approx 0.5\%$

[CLO-02] (06)